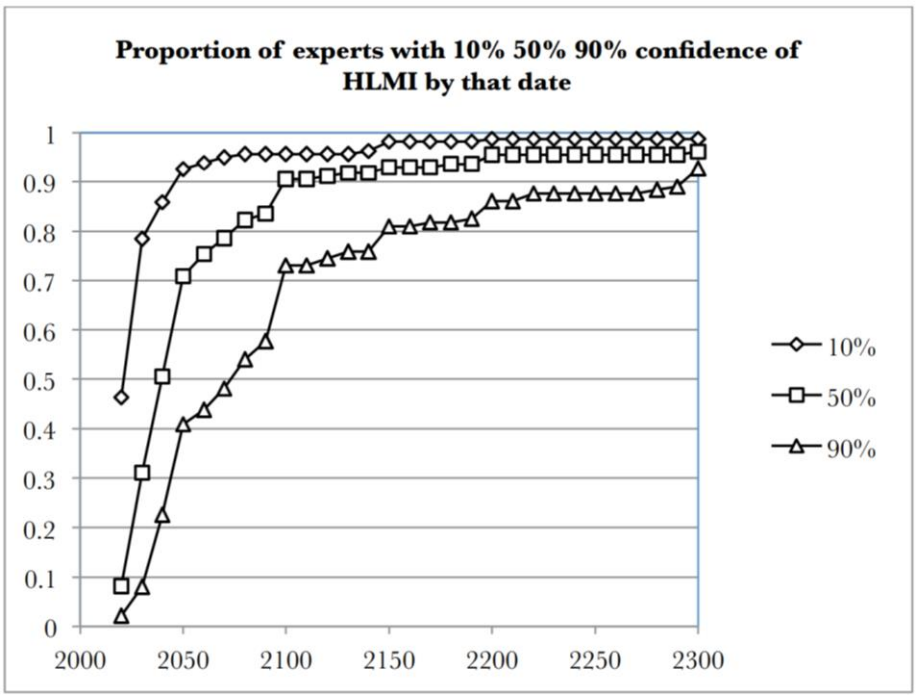
Topic 4: Ethics and Intelligent Computing

What is the ethical way forward for facing the challenges AI poses to human work? resistance, revolution, or somewhere in between?

Introduction

AI presents the world with a wide range of possible outcomes in terms of our relationship to work. Previous industrial revolutions have not led to permanent mass unemployment, and may have even served to reduce it (Butler and Buys 2020). The advent of AI challenges the conventional economic framework that governed these revolutions. In most modern economic theories of growth, capital accumulation and technological progress reduces the amount of labour necessary to produce a good or service, but does not eliminate it (Chirwa and Odhiambo 2018). Labour is generally made more productive through its interaction with new capital and technology. The increased wealth from higher productivity serves to keep labour demand high, especially given its enhanced productivity (Müller 2023). Inasmuch as it seeks to replicate human intelligence, AI could prove to be more of a direct substitute to labour, rather than complement it, jeopardising the traditional economic framework.



Müller and Bostrom (2016), p 11/19. High-Level Machine Intelligence (HLMI) is defined as one that can carry out most human professions at least as well as a typical human.

We have three broad options to wrangle with AI and its implications for work: 1) resist advances in AI that could threaten the status quo; 2) embrace but guide advancements in AI – acknowledging that it could be potentially more disruptive than previous industrial revolutions – but endeavouring for human work to remain and disruption be redressed; and 3) embrace AI with a view to making work obsolete, fundamentally altering the human *modus operandi*. This essay argues that the second aim is the most feasible course of action, and not only for this should it be our aim.

The case for the first and third options

Around 47 percent of current jobs are at high risk of automation within the next decade or two by one estimate. Moreover, the jobs at risk skew towards those associated with lower wages and lower levels of educational achievement (Frey and Osborne 2013). Potential inequalities extend beyond the class dimension. On average, men make up 71 percent of applicants for AI jobs in the US, raising the potential that the gains of the AI revolution emphasise existing disparities in gendered pay differences as AI jobs increasingly become a source of wealth (Shoham et al. 2018). Frey therefore worries that the threat of AI disruption will lead governments to hold back its development, and that this is a genuine possibility, with governments having held back the first industrial revolution (Mokyr 2019).

Holding back technological development leads to less aggregate societal wealth as it is understood within the economics profession today (Chirwa and Odhiambo 2018). Similarly, a technologically advanced but idle world is unlikely to maximise material wealth relative to a world that embraces both technological advancement and human input. A simple consequentialist argument is thus not the strongest proponent of these options, especially since some consequentialists promote the idea of "higher pleasures" where the engagement of our higher faculties (such as scientific inquiry or professional accomplishment) is incommensurately greater than more base forms of pleasure (Mill 1861). These higher pleasures being more absent from a world reached through pursuit of the third option.

Instead, the strongest case for the opposition can be made on distributional grounds. The disruption caused by the AI revolution could lead the least well-off in society to be worse off than if AI developments were restrained, especially given how the jobs at risk skew towards those of the currently least well-off. Were we placed behind Rawls' veil of ignorance, we would therefore opt for a society that held back AI, as this would maximise the welfare of the least well-off (Rawls 1971). Proponents of the third option could also take this line. Striving for a world without work is striving for a work without work-related inequalities. The social makeup of such a world would not value employability, and is therefore likely to be better orientated for the benefit of those that currently experience demerit on the basis of their scarce abilities in the labour market.

The case for the second option and our responsibility in achieving it

Even granting that almost half of current employment is at high risk of automation, the preponderance of employment in the general population is likely to still prevail. Messrs Frey and Osborne themselves conclude that low-skilled workers will reallocate to other tasks, though they will have to acquire new skills to do so with a high degree of success (Frey and Osbourne 2013). Our experience of automation so far is that the indirect effects of interindustry demand linkages, between-industry compositional change, and increases in final demand, more than offset the direct labour-saving effect of automation in advancing industries (Autor and Salomons 2018). The assumption that AI's approximation of human intelligence is the distinguishing feature that will lead this industrial revolution to augur mass-unemployment is not just yet to be proven; it is contrary to the evidence from the closest analogue experiment: immigration. Humans possess human intelligence; they do not merely approximate it, and AI can never be a more complete substitute for humans than other humans. Nonetheless, mass influxes of new workers do not displace current workers on an aggregate level. The Mariel boatlift provides the strongest natural experiment to substantiate this claim. This event refers to the mass influx of immigrants from Cuba to the United States within a six-month period, especially to Miami, increasing Miami's labour force by 7%. The increase of the low skilled segment of the workforce was even more dramatic, since many of the

immigrants were low-skilled. Nevertheless, the Mariel influx had virtually no effect on the wages or unemployment rates of less-skilled Miami workers, even among Cubans who had immigrated earlier (Card 1989). Concentrations and availability of human capital – human intelligence – are positively associated with productivity (Baptista 2003) as the interaction between and cooperation amongst human intelligence raises the productivity of humans as a whole. This evidence of the labour-preserving nature of increases in concentrations of human intelligence favours the view that even if AI truly does approach human levels of intelligence, work will still be ubiquitous as with previous industrial revolutions. *A fortiori*, AI is likely to retain distinct characteristics of intelligence relative to humans, implying greater gains for human interaction with AI as human and artificial intelligence complement each other's blind spots.

The immediately preceding arguments suggest that the natural outcome of the AI revolution will retain human labour, making the third option infeasible unless we are forced into idleness by coercion or a dramatic shift in attitude. Though Frey argues that governments have successfully held back technological developments in the pre-industrial area, the degree of international linkages and competition in the modern era detracts from the feasibility of a successful resistance to advancements in AI. It is far more difficult to prevent the transfer of technology between countries relative to the pre-industrial era, meaning that resistance would have to be universal to be effective. The current geopolitical climate suggests the opposite: government are competing to nurture AI for strategic advantage. China, the US, and Canada have already set out a national AI strategy (Ding 2018), and the Biden administration has restricted China's access to advanced microchips to help maintain the US's lead in AI.

The infeasibility of the first and third options makes it unlikely their pursuit would be consistent with Rawls' difference principle. Resisting an inevitable change sacrifices resources that could be better allocated to the least well-off. Similarly, the existence of work-related inequalities does not itself contradict the difference principle, and the greater societal resources resulting from the acceptance of work can be directed towards improving the wellbeing of the least well-off. The third option also lends itself to the problem of control. Even if AI's interests could be aligned with our own, the degree of control that AI would need to eliminate human work gives us cause to doubt whether any of us (including the least well-off) would truly be better off. The contradiction between such control and human well-being, and our resistance to it, is a feature of venerated literary works such as Aldus Huxley's *Brave New World* and Fyodor Dostoyevsky's *Notes from Underground*, giving us reason to be reluctant to pursue the unprecedented and potentially tyrannical coercion necessary to eliminate work.

Rawls' principle itself is open to criticism, to the benefit of the second option. The difference principle begs the question regarding its implicit assumption that, behind the veil of ignorance, we would be concerned primarily with the distribution of material goods, rather than equally concerning ourselves with what rules would prevail in a just society, or which virtues should be nurtured. Being focused merely on the enjoyment of material goods is ethically myopic, and draws too heavily on consequentialist underpinnings. Orwell exposes how this focus misses much that is of moral concern, recounting how the British working class of the 1930s had an instinctual reluctance to embrace socialism, which purported to benefit them the most in a material sense, despite their economic destitution. The reluctance stemmed from an association, at the time, between socialism and a sort of techno-utopianism in which work was obsolete and effort optional. Yet, "The truth is that many of the qualities we admire in human beings can only function in opposition to some kind of disaster, pain, or difficulty" (Orwell 1937, p. 175). Orwell's account is one that acknowledges virtue, and even within a consequentialist framework, Mill's concept of higher pleasures tries to augment

consequentialism with a concept that approaches that of virtue. Striving to create true AI is a pursuit of excellence, and a society that enables the pursuit is one that nurtures *arête*, especially when it is tempered with efforts to moderate and guide the developments, wisely preserving the challenge of work for society as a whole. Focusing on the rules of a just society, we would also likely rule out the coercion that would be necessary to either prevent the development of AI as with the first option, or to remove work from society as with the third option. Such coercion would strike one as too draconian from behind the veil of ignorance, even if it were to the detriment of distributional aims. The second option is therefore more aligned with an unbiased deliberation behind the veil of ignorance, as it represents a more balanced consideration of the moral ingredients of a just society.

We have ample opportunity to guide the development of AI. General purpose technologies only start becoming transformational when firms reinvent their processes and business models, new products and markets are created, and appropriate skills are acquired (Brynjolfsson, Rock, and Syverson 2018). This gives us the time and the opening to effectively implement the second option. Responsibility for this rests with employers, employees, and governments. Without sacrificing gains in productivity, employers must raise their employees' human capital so that they can fit within a business model that embraces AI. It is incumbent upon employees to anticipate the skills that they must develop to stay on top of their field, or transition to one that will remain relevant. Governments must properly align incentives, ensuring that employers and employees follow through on their responsibilities. The government has a wide range of levers to achieve this, with taxation and regulation being prime examples. Labour suffers a substantial tax burden in many countries, and an informed shifting of this burden on to the shoulders of capital could help reduce distortionary incentives against employing humans. Similarly, ensuring that human creators are remunerated for their output's being used as training for AI would ensure the cost of AI aligns with the true cost of its inputs. Redressing the distributional effects of AI's disruption is also a primary responsibility for governments. The net-gain in resources, responsibly distributed, allows those suffering loss of employment the breathing space to find new ways to add value to society.

Conclusion

This essay has set out the challenge AI poses to humanity and its relationship to work, exploring three normative goals that society could pursue. The second of these goals – a measured approach that embraces but guides the development of AI to work in harmony with human labour – is argued to be the most ethical option, instead of either resisting the development of AI, or promoting a complete revolution that eliminates work with the aid of AI. The second option is presented as the most feasible, making it also consistent with Rawls' difference principle, which is presented as the strongest argument that could be made for the other two options. *A fortiori*, the difference principle is argued to be too narrow-minded, and a more comprehensive set of considerations behind the veil of ignorance would only serve to further support the second option. Finally, the essay sets out the responsibility various stakeholders have in realising the chosen option.

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